the implosion (Chapter 6); led the American atomic bomb project, opposed the hydrogen bomb project early on and then endorsed it and lost his security clearance (Chapter 6); did battle with Wheeler over whether implosion produces black holes (Chapter 6).

Penrose, Roger (b. 1931). British mathematician and theoretical physicist; protege of Sciama's; speculated that black holes lose their hair by radiating it away (Chapter 7); discovered that spinning black holes store huge amounts of energy in the swirl of space outside their horizons and that this energy can be extracted (Chapter 7); developed the concept of a black hole's apparent horizon (Chapters 12 and 13); discovered that the surface areas of black holes must increase, but did not realize the significance of that conclusion (Chapter 12); invented and developed global (topological) methods for analyzing black holes (Chapter 13); proved that black holes must have singularities in their cores and, with Hawking, proved that the big bang contained a singularity (Chapter 13); proposed the cosmic censorship conjecture, that the laws of physics prevent naked singularities from forming in our Universe (Chapter 13).

Press, William H. (b. 1948). American theoretical physicist and astrophysicist; student of Thorne's; with Teukolsky, proved that black holes are stable against small perturbations (Chapters 7 and 12); discovered that black holes can pulsate (Chapter 7); organized the funeral for the golden age of black-hole research (Chapter 7).

Price, Richard H. (b. 1943). American theoretical physicist and astrophysicist; student of Thorne's; gave the definitive proof that a black hole loses its hair by radiating the hair away and proved that anything which can be radiated will be radiated away completely (Chapter 7); saw evidence that black holes pulsate but did not recognize its significance (Chapter 7); with others developed the membrane paradigm for black holes (Chapter 11); worried about Thorne's sanity when Thorne initiated research on time machines (Chapter 14).

Rees, Martin (b. 1942). British astrophysicist; student of Sciama's; developed models that explain the observed features of binary systems in which a black hole accretes gas from a companion star (Chapter 8); proposed that the giant lobes of a radio galaxy are powered by beams of energy that travel from the galaxy's core to the lobes, and with Blandford developed detailed models for the beams (Chapter 9); with Blandford and